

CHASHMAN

Mechatronics & Control Engineer

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PROFESSIONAL SUMMARY

Mechatronics & Control Engineering graduate from UET Lahore with hands-on expertise in PLC programming (Ladder Logic & FBD), HMI design, and SCADA systems. Currently working as a Graduate Engineer Trainee at Pakistan Railways on a real-time interlocking signaling system using TIA Portal and Siemens PLCs. Experienced in industrial automation, sensor integration, safety interlocks, and pneumatic/hydraulic systems through academic projects and professional roles. Also brings teaching experience in PLC programming and industrial automation.

EDUCATION

BE Mechatronics & Control Engineering — UET Lahore

Nov 2021 – Jun 2025

- **CGPA: 3.26 / 4.00** | HEC Recognized | PEC Accredited
- **Coursework:** Control Systems, Industrial Automation & PLCs, Instrumentation & Measurement, Hydraulics & Pneumatics, Embedded Systems, Process Control, Computer-Aided Design

TECHNICAL SKILLS

Industrial Automation & Control: PLC Programming (Ladder Logic, Function Block Diagrams), HMI Design & Configuration, SCADA Systems & Real-Time Monitoring, DCS Fundamentals, Process Control & Optimization, Safety Interlocks, Alarm Management

Instrumentation & Mechanical: Sensors & Transducers, Actuators, Pneumatic & Hydraulic Systems, Control Valves, Conveyor Systems, Flow/Pressure/Temperature Measurement

Software & Programming: Python, C, MATLAB/Simulink, Machine Learning, Data Analysis, SolidWorks (3D CAD), AutoCAD (2D), P&ID Reading

PROFESSIONAL EXPERIENCE

Graduate Engineer Trainee (GET) — Automation & Control — Pakistan Railways, Signal Workshops, Lahore

Apr 2026 – Present

- Working on the PLC-based Tharparkar Railway Signaling Project — a real-time interlocking control system; programmed and debugged ladder logic and FBDs for lever positions, track circuits, point & lock interfaces, and signal feedback networks across Systems A & B in TIA Portal
- Designed an operator HMI dashboard in EasyBuilder Pro with 40 lever controls and simulation inputs for both systems; implemented mutex logic to prevent simultaneous conflicting lever states
- Implemented safety relay (SR) blocks with ON/OFF timers for home, starter, and line signals; integrated SR logic into interface modules and validated all conditions via the HMI dashboard
- Configured Siemens LOGO! 8 PLCs for FBD, ladder logic, and multi-PLC communication over PoE Ethernet; resolved IP conflicts and established cross-PLC I/O communication to simulate networked industrial control

Robotics & AI Instructor — Punjab Group of Colleges

Aug 2025 – Jan 2026

- Delivered instruction in robotics, AI fundamentals, and microcontroller-based systems to engineering students
- Developed structured curriculum and lab exercises covering sensor integration, robot kinematics, and autonomous system design
- Mentored students on hands-on projects involving robot prototyping, embedded programming, and AI-based applications

Research Intern — Machine Learning — NUST, Islamabad

Jun – Jul 2023

- Conducted ML research on brain tumor segmentation using BraTS'23 dataset with data preprocessing, model training, and evaluation
- Contributed to academic publication with detailed technical documentation and reproducible results

STEM & Robotics Instructor — MechaBotix

Jun 2022

- Taught fundamentals of robotics, sensors, and automation; guided prototype development using microcontrollers and sensor modules

ENGINEERING PROJECTS

Automatic Traffic Light Control System (PLC/SCADA)

Nov – Dec 2024

- Programmed PLC using Function Block Diagrams for automated control with normal and emergency operating modes
- Designed HMI for real-time status display and operator interaction with mode switching capabilities
- Integrated SCADA monitoring for centralized visualization, data logging, trend analysis, and alarm management

Automatic Bottle Repositioning System (Hydraulics/Pneumatics)

Nov – Dec 2023

- Designed actuator-sensor logic for automated repositioning on industrial conveyor belts using pneumatic sequencing
- Implemented safety interlocking, sensor-based position feedback, and fail-safe mechanisms for reliable continuous operation

Human-Aware Mobile Robot Navigation (ML/Computer Vision)

May 2024 – Jul 2025

- Developed ML-based autonomous system with real-time sensor fusion and obstacle avoidance achieving 90% navigation accuracy
- Integrated computer vision algorithms for dynamic environment monitoring and adaptive decision-making

LEADERSHIP & ACTIVITIES

- **PR Secretary**, Mechatronics Club, UET Lahore (2024–2025) — Managed event communication, outreach, and student engagement for technical exhibitions and workshops
- **Director Technical**, Mechatronics Club, UET Lahore (2023–2024) — Led project demonstrations at LSM Exhibition'23, YES Week'23, and Trust School Exhibition'23; coordinated technical teams